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8. (a) Aluminium reacts with oxygen to form aluminium oxide.

Using outer electrons only, draw a dot and cross diagram to show the bonding in aluminium oxide. [2]

(b) Aluminium also reacts with chlorine to form aluminium chloride.

A 0.400 g sample of aluminium chloride was heated to 220 °C. The vapour produced occupied a volume of 60.8 cm^3 at a pressure of 101 kPa.

Show that the molecular formula of aluminium chloride in the vapour is Al_2Cl_6. [4]

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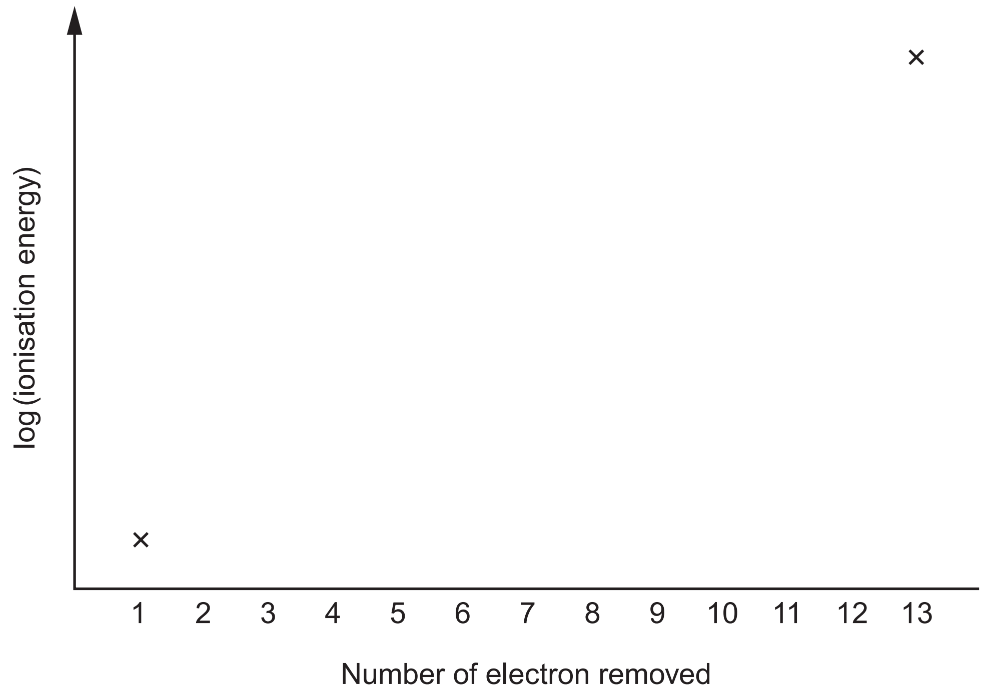
(c) (i) Sketch a graph of log (ionisation energy) for the successive ionisations of aluminium.

The first and last points have been plotted for you. [1]



(ii) I. Explain the general slope of the graph. [1]

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II. Explain the reason for any sharp changes in the graph. [1]

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(d) Although aluminium has 25 known isotopes only two of them occur naturally. These are ^{27}Al which is stable and ^{26}Al which is radioactive.

^{26}Al decays by electron capture and its half-life is 7.2×10^5 years.

(i) Give the mass number and symbol of the species produced when an atom of ^{26}Al decays. [1]

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(ii) If 8.0 mg of ^{26}Al decays by electron capture, calculate the mass in **grams** of ^{26}Al left after 2.88×10^6 years. [2]

Mass = ... g

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